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Εθνικόν και Καποδιστριακόν
Πανεπιστήμιον Αθηνών
— ΙΔΡΥΘΕΝ ΤΟ 1837 —

MSc in Computer Science

Data, Information, and Knowledge Management

Data Visualization (M126)

Spring Semester 2025-26

Group project proposal

Hormuz Under Pressure: Tanker Risk and Greek Fuel Prices

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Abstract

This project presents an interactive visual story about how risk at the Strait of Hormuz concentrates global energy exposure and connects to everyday fuel prices in Greece. The website moves from the narrow maritime chokepoint, to global tanker traffic, vessel composition, oil-flow destinations, global fuel benchmarks, and Greek pump prices by prefecture. The final implementation is a static D3 and Leaflet web project designed for a broad non-specialist audience.

Background and Motivation

The Strait of Hormuz is a narrow passage between Iran, Oman, and the United Arab Emirates. It is one of the most important maritime energy chokepoints because large volumes of crude oil, condensate, petroleum products, and LNG pass through it. In the project narrative, Hormuz is treated as a place where a local disruption can have global consequences because energy flows are geographically concentrated.

The final website uses a disruption marker called HORMUZ-26 to separate baseline and post-shock periods. Instead of claiming a simple direct causality, the story visualizes a propagation chain: a chokepoint shock affects tanker traffic, benchmark fuel markets react, and the user then examines how Greek retail fuel prices changed nationally and across prefectures.

Our motivation is to make an abstract geopolitical and energy-security issue visible through linked, interactive views. The dashboard avoids presenting the Strait of Hormuz as a distant news item only. It shows why the chokepoint matters, which oil flows depend on it, and how users in Greece can observe fuel-price differences across regions and fuel types.

Project Aims and Research Questions

The main aim is to build a polished web-based data visualization project that explains the energy-risk chain from maritime chokepoint to Greek fuel prices. The final story is organized into linked chapters that combine maps, rankings, time series, vessel-mix charts, flow charts, and a Greek prefecture map.

Key questions:

1. Where is the Strait of Hormuz, and why does its geography make it a critical energy chokepoint?
2. How does Hormuz compare with other maritime chokepoints in tanker volume and tanker dependence?
3. What types of ships pass through Hormuz, and how did vessel traffic change after the HORMUZ-26 disruption marker?

4. Which countries send oil through Hormuz, and which destination markets are most exposed to those flows?
5. How did Brent crude, jet fuel, and Greek retail fuel prices change after the disruption marker, nationally and by prefecture?

Target Audience

The target audience is broad and non-specialist: students of data visualization, readers interested in energy and geopolitics, Greek consumers who want to understand fuel-price differences, and journalists or instructors looking for a clear visual explanation of energy chokepoint risk. The website is designed to work as a guided story, while still allowing exploration through filters, hover tooltips, linked highlighting, search, and full-screen maps.

Datasets Considered

Primary datasets used in the final implementation:

- **IMF PortWatch maritime chokepoint data.** Daily chokepoint traffic is used to compare Hormuz with 27 other chokepoints and to build the vessel-class chapter for Hormuz. The final website uses tanker counts, total transits, tanker share, capacity, and vessel classes such as tankers, containers, dry bulk, general cargo, and Ro-Ro.

Source: IMF PortWatch, processed into derived CSV files for the website.

Supplementary and contextual datasets:

- **U.S. Energy Information Administration, Strait of Hormuz analysis.** Used for the introductory context panel and for the main energy-security figures in the narrative, including 2024 oil flows through Hormuz and the importance of Asian destination markets.

<https://www.eia.gov/todayinenergy/detail.php?id=65504>

- **EIA figure data based on Vortexa tanker tracking.** A cleaned Excel-derived dataset is used for origin countries through Hormuz and top destination markets. The charts separate named countries from grouped regions where the source uses aggregation.

Underlying source cited by EIA: Vortexa tanker tracking; cleaned dataset derived from EIA figure data.

- **FRED, Brent Crude Oil Price (DCOILBRETEU).** Used for the global benchmark time-series view that connects the maritime disruption marker to oil-market movement.

<https://fred.stlouisfed.org/series/DCOILBRETEU>

- **FRED / U.S. EIA, Kerosene-Type Jet Fuel Prices, U.S. Gulf Coast (DJFUELUSGULF).** Used as a fuel-market benchmark alongside Brent crude.

<https://fred.stlouisfed.org/series/DJFUELUSGULF>

- **Greek fuel price data from fuelprices.gr and the Hellenic Ministry of Development.** Daily fuel-price records are used for national price lines and for prefecture-level maps and rankings.
<https://www.fuelprices.gr/>
- **Greek prefecture boundaries and centroid support data.** Used for the choropleth map, linked highlighting, search, and geographic focus behavior in the Greek fuel-price chapter.
Boundary layer based on public GeoJSON data, refined with project-specific name matching for Greek prefectures.
- **OpenStreetMap / Leaflet basemaps.** Used for the global and chokepoint maps where geographic context is needed. The Greek prefecture map uses a simplified geometry-focused view to reduce visual clutter.
<https://leafletjs.com/> and <https://www.openstreetmap.org/>
- **Derived CSV outputs.** The website runs on lightweight processed files stored in the repository, including daily Hormuz traffic, chokepoint summaries, origin and destination flow tables, national Greek fuel prices, prefecture fuel prices, and event markers.

Data processing and cleaning:

- Processing was carried out in Python with pandas. Dates were normalized, missing values were handled, daily records were aggregated where needed, and before/after periods were defined around HORMUZ-26. The Greek fuel data was reshaped into fuel-specific prefecture tables for A95, A100, diesel, heating diesel, and LPG.
Geographic preparation included matching English and Greek prefecture names, preserving island prefecture parts where appropriate, and adding safeguards for linked map and ranking interactions.
- The final website does not rely on server-side computation. All processed data is loaded as static CSV or GeoJSON files by the browser.

Data limitations:

Some source datasets use grouped regions such as Europe or Other Asia rather than individual destination countries. These labels are retained where they exist in the source, and the charts avoid presenting them as individual countries.

The visual story emphasizes observed patterns and before/after comparison windows. All datasets used in the final implementation are real public, official, or source-derived datasets; no synthetic or generated dataset is used. HORMUZ-26 is an analysis marker used to define comparison windows, not a generated data series. The website does not claim to prove a deterministic causal relationship between Hormuz events and every local retail fuel-price movement.

Visualization Design and Interaction

The final website is structured as a visual story in linked chapters, moving the user from the global chokepoint to Greek retail fuel prices:

1. **CH-01, The Narrow Gate:** an annotated map of the Strait of Hormuz and KPI cards explaining why the location matters for global oil and LNG flows.
2. **CH-02, World Comparison:** a linked map and ranking of 28 maritime chokepoints. Users compare before and after HORMUZ-26 by tankers per day and tankers as a percentage of total traffic. Selecting a chokepoint highlights it both on the map and in the ranking, with a clear reset option.
3. **CH-03, Vessel Mix:** small-multiple time-series panels and a donut summary showing which vessel classes pass through Hormuz and how they changed after the disruption marker.
4. **CH-04, Oil Flows:** an origin-country map and paired bar charts showing countries sending oil through Hormuz and the top destination markets according to the EIA/Vortexa-derived data.
5. **CH-05, Greek Prices:** national fuel-price lines for Greece, followed by a prefecture map and dynamic ranking for A95, A100, diesel, heating diesel, and LPG. Users can switch between latest price and change since HORMUZ-26.
6. **Video and Credits:** the website includes a short explainer video section and a credits/source section documenting the datasets and tools used.

Interactions include hover tooltips, linked map and chart highlighting, period and metric filters, fuel-type filters, dynamic search for Greek prefectures, full-screen map buttons, and reset controls to remove selections.

Significance and Visual Data Science Challenges

The significance of the project lies in connecting a global maritime chokepoint to a concrete local consumer issue: fuel prices in Greece. The website shows how a location that most users may never see can structure global oil flows, tanker traffic, benchmark prices, and regional price differences.

- **Multi-scale storytelling:** the project moves from a single narrow strait, to global chokepoints, to origin and destination countries, to national Greek prices, and finally to individual prefectures.
- **Mixed data types:** the website combines geospatial boundaries, daily time series, ranking tables, vessel-class counts, and country-flow data. The challenge is to keep the story coherent without overwhelming the user.

- **Linked interaction design:** selecting a chokepoint or Greek prefecture must update related views without breaking context. The final implementation uses linked highlighting, dynamic list additions, and reset controls to keep interactions understandable.
- **Visual clarity and avoiding overclaiming:** the project shows before and after patterns around HORMUZ-26 but avoids presenting local fuel-price changes as monocausal. Text labels and source notes are used to keep the interpretation transparent.
- **Geographic accuracy:** Greek prefectures with islands and multi-part geometries required careful matching so that map highlights remain consistent with the ranking and search results.

Tools and Implementation

The website is implemented as a static front-end project using HTML, CSS, vanilla JavaScript, D3.js, and Leaflet. Python and pandas were used for data processing and export to lightweight CSV/GeoJSON files. The final site can be hosted on GitHub Pages or any static web server. D3 is used for charts and number formatting, while Leaflet is used for interactive maps.

Must-have Features

- Guided chapter structure from Hormuz geography to Greek fuel prices
- Linked world chokepoint map and comparison ranking with before/after period filters
- Vessel-mix time series and composition summary for Hormuz traffic
- Origin and destination oil-flow views based on EIA/Vortexa-derived data
- Greek fuel-price chapter with national lines, prefecture choropleth, dynamic ranking, search, and fuel-type filters
- Static public website with video section, credits page, and documented data sources

Optional Features

Optional extensions include adding more European fuel-price comparisons, expanding the destination-flow layer with additional tanker-tracking data, adding an uncertainty note panel, and building a compact mobile-first version of the story.

Schedule

Final implementation timeline, targeting submission on 20 May 2026.

Weeks 1 to 2 (27 April to 10 May):

Dataset collection, source validation, cleaning, exploratory analysis, and early visual experiments. The work used only real public or official datasets and defining the final Hormuz-to-Greece story structure.

Weeks 3 to 4 (11 May to 17 May):

D3 and Leaflet prototyping, chapter integration, map interactions, period and metric filters, source notes, and visual style alignment across the website.

Week 5 (18 May to 19 May):

Final website polish, Greek prefecture map corrections, proposal and README preparation, video section setup, GitHub packaging, and final source documentation.

Final day (20 May):

Deployment checks, final documentation review, and submission.